

CBSE Class 10 Mathematics - Answer Key

1. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x and y are prime numbers, then the $HCF(a, b)$ is:

Answer: (B)

Solution:

1. Find the common prime factors with their lowest exponents.
2. The prime factors of a are x, x, x, y, y .
3. The prime factors of b are x, y, y, y .
4. The common factors are one x and two y s, i.e., xy^2 .

Derivation:

$$\begin{aligned}a &= x^3y^2 \\b &= xy^3 \\HCF(a, b) &= x^{\min(3,1)} \times y^{\min(2,3)} = xy^2\end{aligned}$$

Real Numbers — Fundamental Theorem of Arithmetic

2. The decimal expansion of the rational number $\frac{43}{2^4 \times 5^3}$ will terminate after how many places of decimals?

Answer: (D)

Solution:

1. To terminate, the denominator must be in the form $2^n \times 5^m$.
2. The number of decimal places is $\max(n, m)$.
3. Here $n = 4$ and $m = 3$.
4. Therefore, it terminates after 4 places.

Derivation:

$$\frac{43}{2^4 \times 5^3} = \frac{43 \times 5}{2^4 \times 5^4} = \frac{215}{10^4} = 0.0215$$

Real Numbers — Rational Numbers and Decimal Expansions

3. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is:

Answer: (C)

Solution:

1. Find the HCF of 65 and 117 using prime factorization.
2. $65 = 5 \times 13$, $117 = 3^2 \times 13$. HCF is 13.

3. Set $65m - 117 = 13$ and solve for m .

Derivation:

$$65m - 117 = 13$$

$$65m = 130$$

$$m = 2$$

Real Numbers — Euclid's Division Lemma

4. The largest number which divides 70 and 125 leaving remainders 5 and 8 respectively is:

Answer: (A)

Solution:

1. Subtract the remainders from the numbers: $70 - 5 = 65$ and $125 - 8 = 117$.
2. Find the HCF of 65 and 117.
3. $65 = 5 \times 13$ and $117 = 9 \times 13$. HCF = 13.

Derivation:

$$70 - 5 = 65$$

$$125 - 8 = 117$$

$$HCF(65, 117) = 13$$

Real Numbers — HCF and LCM Applications

5. If p and q are two prime numbers, then their LCM is:

Answer: (D)

Solution:

1. For any two prime numbers, they have no common factors other than 1.
2. The LCM of two co-prime numbers is their product.

Derivation:

$$LCM(p, q) = \frac{p \times q}{HCF(p, q)} = \frac{p \times q}{1} = pq$$

Real Numbers — Fundamental Theorem of Arithmetic

6. Which of the following is a non-terminating repeating decimal?

Answer: (C)

Solution:

1. A rational number is non-terminating repeating if its denominator prime factorization contains factors other than 2 or 5.
2. Check denominators: $8 = 2^3$, $16 = 2^4$, $15 = 3 \times 5$, $10 = 2 \times 5$.

Derivation:

$$\frac{7}{15} = \frac{7}{3 \times 5}. \text{ Since denominator has 3, it is non-terminating repeating.}$$

Real Numbers — Rational Numbers and Decimal Expansions

7. The product of a non-zero rational and an irrational number is always:

Answer: (B)

Solution:

1. Consider a rational $r \neq 0$ and irrational i .
2. If ri were rational, then i would be the ratio of two rationals, which is rational, a contradiction.

Derivation:

$$\text{Let } r = \frac{p}{q}, i \text{ be irrational. } r \times i = k$$

$$\Rightarrow i = \frac{kq}{p}$$

$\Rightarrow i$ is rational, contradiction.

Real Numbers — Irrational Numbers

8. If n is a natural number, then $9^n - 4^n$ is always divisible by:

Answer: (C)

Solution:

1. Test for $n = 1$: $9^1 - 4^1 = 5$.
2. Test for $n = 2$: $9^2 - 4^2 = 81 - 16 = 65$.
3. Since 5 and 65 are both divisible by 5, the expression is divisible by 5.

Derivation:

$$\begin{aligned} a^n - b^n &\text{ is divisible by } (a - b). \\ 9^n - 4^n &\text{ is divisible by } (9 - 4) = 5. \end{aligned}$$

Real Numbers — Fundamental Theorem of Arithmetic

9. The value of the exponent of 2 in the prime factorization of 144 is:

Answer: (D)

Solution:

1. Perform prime factorization of 144.
2. $144 = 12 \times 12 = (2^2 \times 3) \times (2^2 \times 3) = 2^4 \times 3^2$.
3. The exponent of 2 is 4.

Derivation:

$$144 = 2 \times 72 = 2^2 \times 36 = 2^3 \times 18 = 2^4 \times 9 = 2^4 \times 3^2$$

Real Numbers — Fundamental Theorem of Arithmetic

10. Given that $HCF(252, 594) = 18$, the $LCM(252, 594)$ is:

Answer: (B)

Solution:

1. Use the formula $HCF \times LCM = a \times b$.
2. $18 \times LCM = 252 \times 594$.
3. $LCM = \frac{252 \times 594}{18} = 14 \times 594 = 8316$.

Derivation:

$$LCM = \frac{252 \times 594}{18} = 14 \times 594 = 8316$$

Real Numbers — HCF and LCM Applications